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VisualSiteDiary: A detector-free Vision-Language Transformer model for captioning photologs for daily construction reporting and image retrievals

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ABSTRACT

This paper presents VisualSiteDiary, a Vision Transformer-based image captioning model which creates human-readable captions for daily progress and work activity log, and enhances image retrieval tasks. As a model for deciphering construction photologs, VisualSiteDiary incorporates pseudo-region features, utilizes high-level knowledge in pretraining, and fine-tunes for diverse captioning styles. To validate VisualSiteDiary, a new image captioning dataset, VSD, is presented. This dataset includes many realistic yet challenging cases commonly observed in commercial building projects. Experimental results using five different metrics demonstrate that VisualSiteDiary provides superior-quality captions compared to the state-of-the-art image captioning models. Excluding the task of object recognition, the presented model also outperformed mPLUG –the state-of-the-art visual-language model– in the image retrieval task by 0.6% in precision and 0.9% in recall, respectively. Detailed discussions illustrate practical examples on how VisualSiteDiary improves the process of creating daily construction reports, paving the way for future developments in the field.

1. Introduction

Construction sites are a hub of a variety of work activities conducted by different trade contractors. For site managers, keeping everything in check across all work areas, poses a considerable challenge. While various tools provide a general project overview and expected progress, there remains one contractually-required practical tool that reveals the true, real-time progress: daily logs or Daily Construction Reports (DCRs). These DCR logs are a living record of a project as they represent the details of each day's progress. Specifically, DCRs track on-site activities, keeping stakeholders on the same page on 'what is there' vs. 'what should be there', improving productivity and minimizing delays [1]. Neglecting or the absence of proper DCR documentation heightens the risk of a minor problem escalating into a major one, as DCRs are critical for protecting against disputes. Despite the significant value of DCRs, collecting, analyzing and reporting them is associated with several challenges:

- *Collecting DCRs is time-consuming*– To create DCRs, a superintendent is required to take dedicated notes throughout the day. Alternatively, one or two field engineers are put in charge of collecting, analyzing, and tabulating notes provided by all trade contractors. Because the work is time-consuming [2–5], not a whole lot of meaningful information is captured in these DCRs. For example, a subcontractor may simply report "framing" work, without documenting where, when, what, and how much work happened. Often, there are photos taken throughout the day, nevertheless, those photos are not closely correlated with the DCRs, causing the wealth of information captured in photos to also remain untouched.
- *Analyzing DCRs is a data-intensive process*– Tabulating non-descriptive written note in plain English as well as correlating them with pictures is a daunting task. As such, DCRs are not properly used to prioritize, manage, and track work activities often. Instead, receiving progress updates is mainly conducted through phone calls or it is left for in person meetings in the trailer [1,6,7].
- *DCRs in their current form captured across many projects are not actionable*– Because of the challenges associated with collecting and analyzing photos and daily notes and the reliability issues of DCRs, critical discussions around delays and schedule revisions are delayed until weekly coordination meetings, which is often too late to proactively address potential delays [2,7–9].

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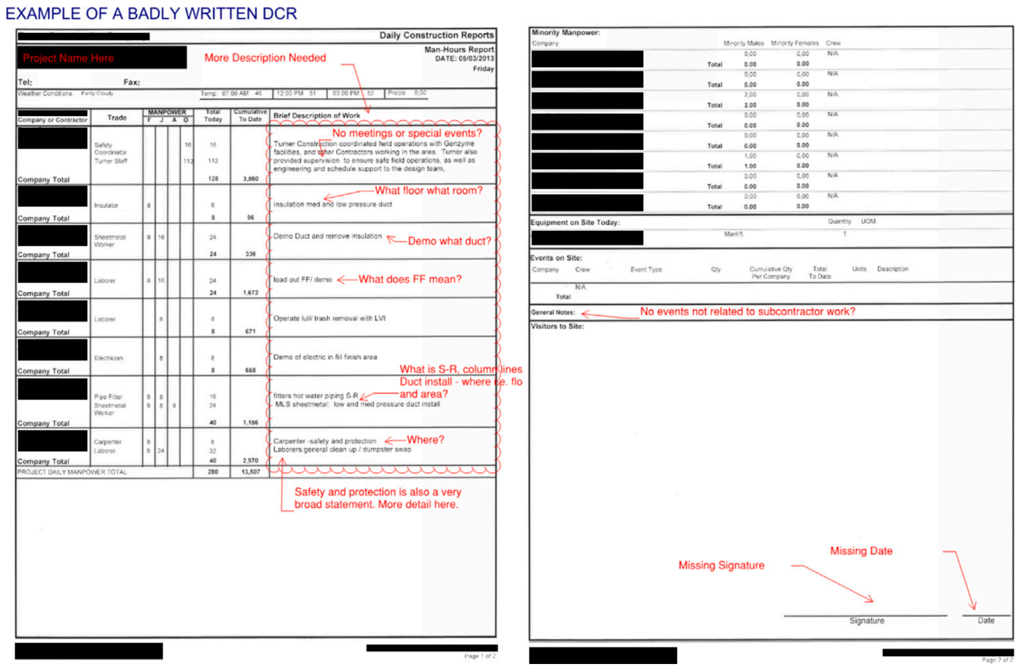

Fig. 1. A real-world example of a Daily Construction Report (DCR) that exhibits poor quality.

Fig. 1 illustrates an example of a DCR that lacks actionable insights around "where", "how" and "what" of the work that happened on-site on a particular calendar date.

A separate body of computer vision techniques has been developed to analyze images and videos for identifying progress changes or quality issues, or detecting, tracking, and analyzing actions and the work context of resources –i.e., labor, equipment, and material– on project sites. A thorough review of these works can be found in [10,11]. To date, these techniques have not extended to infer human-readable descriptions of work activities (i.e., "what" work happened on the site, and or "how" the site has evolved over time), which is necessary for DCR reports. Most recent works have focused on image captioning techniques to generate human-readable descriptions for construction images [12–14]. However, the produced descriptions to date, do not offer any insights on the "what" and "how" of the work and specifically what objects the described work has been applied to. Their datasets also do not cover diversity of construction activities that are typically found on a construction project. Furthermore, to mitigate the manual drafting of documents such as reports, meeting agendas, and change requests, pre-defined templates or ontologies are developed for various document types [15]. Nonetheless, this approach remains inflexible and still necessitates manual effort to populate content and format it into human-readable descriptions.

To fill in these gaps, this paper introduces a new vision transformer-based model, called VisualSiteDiary, which builds on a pretrained large vision-language model. VisualSiteDiary aims to automatically generate image content descriptions on various scenarios, enabling (1) automated creation of construction daily reports and (2) image retrieval using text inputs from users. To provide essential insights required for DCRs, we present the VSD image-caption dataset, which comprehensively deciphers the "what" and "how" of the work, along with the types of elements involved in construction activities, for the generation of detailed work descriptions for DCRs. Fig. 2 shows how VisualSiteDiary enables these use-cases.

To improve the quality of text-based image descriptions, VisualSiteDiary: (1) incorporate pseudo-region features; (2) utilize high-level knowledge for the pre-training step where the model pre-learns overall difficulty knowledge of each input image, and (3) is fine-tuned based on users' preference for image captioning styles. VisualSiteDiary demonstrates that incorporating these three unique approaches improves image captioning capabilities necessary for the identified use-cases. To validate VisualSiteDiary, a new dataset combining four different datasets with new construction activity-oriented captions is provided for DCR generation. The main technical contributions are summarized in the following section.

2. Related work

The advent of deep learning approaches used in natural language processing (NLP) and computer vision (CV) techniques, has transformed the way meaningful information is extracted from commonly available image and text data from construction sites [16,17]. Here, we delve into three key areas as outlined below:

2.1. Reality capture and reality mapping as stepping stones

Over the past decade, reality capture and reality mapping solutions have established themselves as proven techniques for visually capturing and documenting existing conditions on construction. We differentiate these solutions from one another by (a) categorizing *reality capture* as the process of manually or automatically pinning 360-degree or perspective images on a floor plan; typically through Simultaneous Localization and Mapping (SLAM) techniques; and (b) *reality mapping* as the process of creating as-built 3D point clouds, mesh models, and schematic floor plans from 360-degree or perspective videos and images; typically through a pipeline of image-based 3D reconstruction techniques. The result of either one of these two categories of solutions is an image-walk-through experience similar to Google street view. In the case of reality mapping, these image walk-through experiences are measurable. By visually communicating ongoing work and highlighting discrepancies between as-designed and as-built conditions, these solutions have already proven themselves to save time in construction coordination tasks, and reduce project risk [9,18,19]. Reality data offer insights into the chronological "when" (timestamp) and geographical "where" (spatial context) of the ongoing work; however, understanding "what" work happened on the site, and or "how" the site has evolved over time is often left to interpretation of the users.

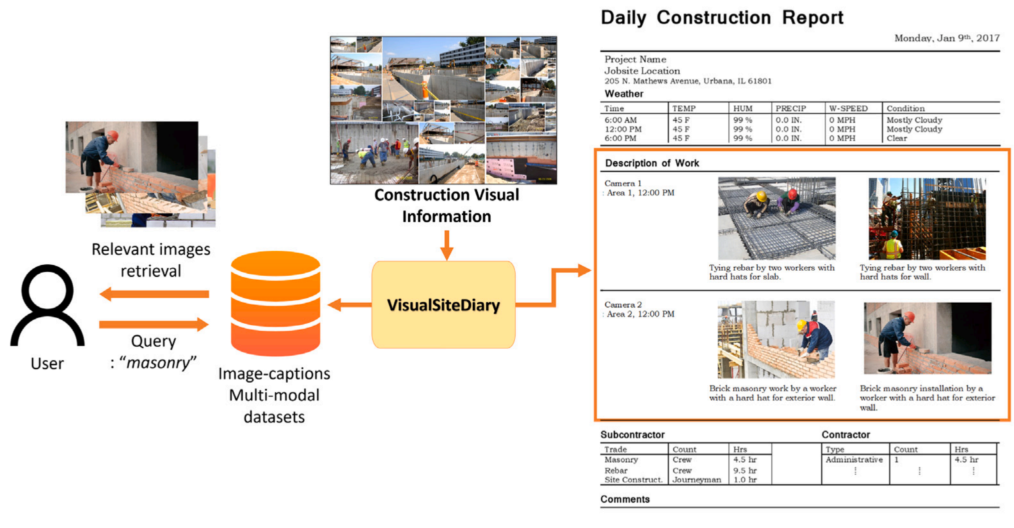

Fig. 2. Overall concepts of automated daily construction reporting and text–image retrieval via VisualSiteDiary.

2.2. Computer vision

Over the past two decades, the availability of large volumes of reality capture data has triggered a large body of computer vision research. Thorough reviews of these techniques can be found in [10,11]. Most relevant examples include: (a) extracting low-level features such as intensity, color distribution, and texture, using filters [20–22] and more recently deep learning networks such as VGG-19 and ResNet-101 [23] for material cluster recognition and image retrieval, (b) 3D as-built modeling and progress monitoring [2,8,24], (c) damage inspection [25–28], and Convolutional Neural Networks (CNNs) such as Faster R-CNN [29] and 3D ResNet for (d) productivity assessments [30–32], and (e) worker safety analysis [33–36] using ⟨Agent – Action – Object⟩ [34] tuples. These works exploit and extract spatio-temporal detection and analysis of objects, and their context. As such, their applications were not extended to the task of providing a human-readable description of the image needed for DCR and information retrieval purposes.

2.3. Natural language processing

Recent research at the interface of NLP and computing in construction has shown promising results in automatically discovering and extracting crucial knowledge to support a series of construction decision-making tasks [17,37,38]. Text mining and NLP techniques have both been used to extract knowledge from *unstructured* and *structured* text data commonly available in change orders, field reports, accident records, and construction plans and schedules to solve for compliance checking [39–41], safety management [42–44], and planning and scheduling [45–49].

For compliance checking, researchers have used Support Vector Machine Learning (ML) algorithms with bag-of-words representations [39], extracted syntactic and semantic features for extending Building Information Modeling (BIM) data with regulatory concepts [40], and extracted rule terms and logic relationships from regulatory documents [41]. For safety management, NLP techniques have been utilized for rule-based keyword matching [42] and ML algorithms based on Term Frequency-Inverse Document Frequency word vectorization [43]. Additionally, Word2Vec embeddings have been used for retrieving relevant safety accident cases from existing reports [44]. In the context of construction planning, NLP techniques have been applied to automatically learn planning knowledge from existing construction schedules. To formalize construction planning and sequencing knowledge, Long Short-Term Memory Recurrent Neural Networks (LSTM-RNNs) are employed to model schedule activity sequences [45,46] and to translate human-written scheduling language to machinery ontological language [47]. For the automated generation of look-ahead plan revisions, a Transformer-based language model is used to augment implicit dependency constraints in schedule sequences [48]. Alternatively, a chatbot system was proposed in [5], allowing construction workers to submit daily construction reports by retrieving stored information based on natural language interaction.

Current techniques mainly focus on generating actionable insights from text documents and as such there were not directly extended to make insightful interpretation from construction photologs or integrating data from both text and image sources [17].

2.4. Image captioning of construction photologs

In recent years, large-scale transformer-based language models such as GPT [50] and BERT [51] have significantly gained attention due to their outstanding performance in various NLP tasks. The success of pretraining these models on large datasets and fine-tuning them for specific downstream tasks inspired researchers to explore their application in computer vision domains, such as image recognition, segmentation, and captioning tasks [52].

Image captioning, mainly enabled through multimodal deep learning, has gained interest within the Architecture/Engineering/Construction (AEC) community [53] as well. To do so, CNN-RNN based architectures are mainly exploited to encode images and decode for generating word sequences [12,14,54–56]. The CNN encoder is used to extract image features as an object detector, and an RNN decoder is used to produce final text-based descriptions. Building on the CNN-RNN-based encoding–decoding architecture, image captioning models are presented to generate construction jobsite activity descriptions [12,14,55], describe worker behavior descriptions [54], and explanations of bridge damage [56]. Building on the successful performance of transformer networks in NLP tasks, a transformer decoder has been used to generate construction heavy equipment scenes by combining a ResNet-101 encoder with self-critical sequence training (SCST) strategy [13]. Using a CNN-based image encoder as an object detector and a language generation decoder, showed promising results in retrieving human-readable descriptions from jobsite images.

While these preliminary works show the potential of integrating NLP and CV techniques, current models are still largely dependent on the performance of a CNN-based object detector. These models incur substantial computational costs through processes like extracting salient regions with bounding boxes and employing regional operations such as Region of Interest Pooling and Region Proposal Network within modern CNN-based networks. These intermediate operations, unfortunately, lead to training inefficiencies and heightened inference latency during predictions [57–60]. In addition, for generating captions, CNNs typically operate locally, processing input images via convolutional operations, while RNN-based networks handle the linguistic part in an end-to-end manner. This architecture can lead to difficulties in effectively modeling relationships between distant visual and linguistic elements, resulting in an incapacity to fully grasp long-range dependences and global context across both modalities [61].

2.5. Detector-free vision-language transformer model for captioning photologs

Recently, the Vision-Language Transformer (VL) architecture, integrating a pre-trained Vision Transformer (ViT) as an image feature encoder and a separate Transformer decoder for text generation, has become popular for image captioning tasks, showing state-of-the-art performance, while eliminating detectors. In general image captioning, such as using MS COCO-caption [62], Mokady et al. [57] utilize a mapping network to feed image embeddings extracted from a vision-language pre-trained Transformer model into a pre-trained Transformer language model for generating image captions. Fang et al. [59] propose a vision transformer-based captioning model incorporating Concept Token Network for semantic prediction at the token level. Li et al. [58] introduce a pre-trained model with cross-modal skip-connections, significantly improving performance across multiple Vision-Language tasks, including image captioning. Moreover, Chen et al. [63] present self-resurrecting activation units for balancing features from the visual and textual modalities, achieving state-of-the-art results on a medical report generation dataset. For a comprehensive understanding of a general Vision-Language Transformer architecture, refer to Section 3.1.

This transition to Vision-Language Transformer models yields significant improvements in both runtime and parameter efficiency [60]. However, flattened grid image features from the Transformer architecture can lead to a loss of spatial information [64]. Furthermore, these general captioning models are solely evaluated on general image-caption datasets. Considering the dynamic and intricate nature of construction field images, these Vision-Language Transformer models tailored for general domains are not specifically designed, trained, or validated for creating DCRs, which require a detailed and complex explanation of "what" and "why" of the ongoing activities tied against "when" and "where" provided through reality capture or reality mapping data.

Also, the construction image-caption dataset used in [12,14,54] only includes pictures from construction activities related to a building superstructure (e.g., masonry, reinforcing steel, and scaffolding work), while Xiao et al. [13] only covers site work or activities related to excavating a substructure which involves heavy equipment. Hence, the performance of these models has only been tested under limited conditions and not comprehensively for typical DCR-related use-cases. Also, their captioning did not include the critically needed information on "what" is happening in the photo. For instance, the captioning in Zhong et al. [14] displays descriptions such as "a worker in a yellow helmet is welding steel", without describing "what" type of work happening and on what element type (e.g., placing rebars for columns), even though the 'how' (welding) and 'who' (a worker) information is included. To offer valuable insights needed for DCRs, it is crucial to fully decipher the type of elements a construction activity applies to, such as those related to 'column', 'deck', 'exterior wall', or 'foundation' for generating DCR work descriptions.

2.6. How gap in knowledge is addressed in this paper

This paper addresses key challenges to facilitate seamless communication between visual and textual information in diverse construction scenarios, leading to automated creation of Daily Construction Reports (DCRs) and reliable image–text retrieval. These challenges, outlined in Section 2.5, can be summarized as follows:

- Lack of detailed descriptions for DCRs: Without specifying "what" type of work and elements involved, hindering the creation of construction activities pertinent to DCRs (e.g., welding steels (others) → placing rebars for columns (ours)).
- Limited diversity in image-text datasets: Existing datasets only cover superstructure activities or earthwork separately, lacking comprehensive representation of various construction scenarios.
- Handling spatial information: Directly processing flattened grid visual features within the Transformer architecture poses a loss of spatial information.
- Understanding construction scene complexity: The absence of pre-trained Vision-Language models capable of understanding the intricacies and difficulties of each construction scene for caption generation.
- Inflexibility of caption generations: Limited to generating captions in a single style, disregarding user preferences and varying use-case requirements.

Different from previous works, we present (1) a new comprehensive visual dataset with ground truth daily reporting descriptions, covering a larger range of construction activities typically found in building construction projects, including site work, substructure, and superstructure-related activities. Also, our ground-truth captions are detailed with each significant work item, 'activity purpose', using '*for what*' phrase (e.g., for site work, for foundation, for exterior wall finish, for slab) to create fully detailed descriptions for DCRs. Next, we present (2) a detector-free Vision-Language Transformer model that generates human-readable descriptions on various construction scenes, which solely builds on a visual transformer architecture [52] instead of relying on an object detector encoder. To further enhance the quality of image descriptions, our model (VisualSiteDiary) is built on a pre-trained large vision language (VL) model (mPLUG) [58], introducing (3) Pseudo-Region (PR) features to mitigate spatial information loss within the Transformer architecture, and (4) High-level image knowledge Pre-training (HP) before fine-tuning to discern the complexity (i.e., difficulty) of each construction scene, enabling the model to allocate greater consideration on images with higher difficulty levels during caption generation. As a result, VisualSiteDiary outperforms the state-of-the-art Vision Transformer baselines on our VSD dataset, generating superior-quality captions. Also, during fine-tuning, VisualSiteDiary (5) generates each different style of captions based on user preferences. A prototype demo to automate the generation of construction image descriptions purely based on visual data; available to the public.[1] The following sections cover details on VisualSiteDiary's architecture, our dataset, experiment results, and discussions.

3. Method: VisualSiteDiary

In this paper, the primary objective is to enable seamless communication between visual and textual information in diverse construction scenarios, resulting in the automated creation of DCRs and reliable image-text retrieval. To achieve this, this paper presents, VisualSiteDiary which builds upon a pretrained vision-language mPLUG model [58], by employing three key strategies: effective visual region feature incorporation, a novel training method based on high-level key information (the level of difficulty of each input image dataset), and fine-tuning to leverage different styles of captions. The following sections explain these three approaches combined and show how VisualSiteDiary effectively generates higher-quality captions reflecting different caption styles (see Fig. 3).

[1] Our new caption data and code are made available at https://github.com/joonv2/VisualSiteDiary.

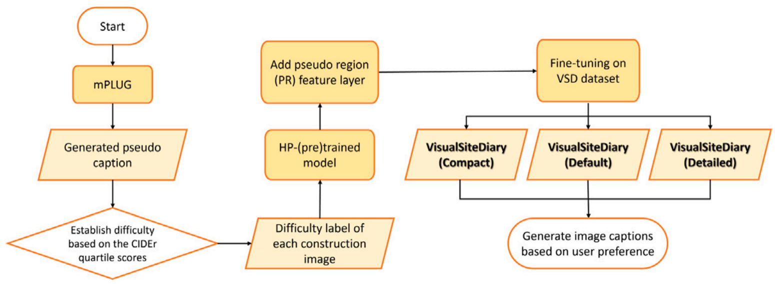

Fig. 3. The overall flowchart of our proposed VisualSiteDiary.

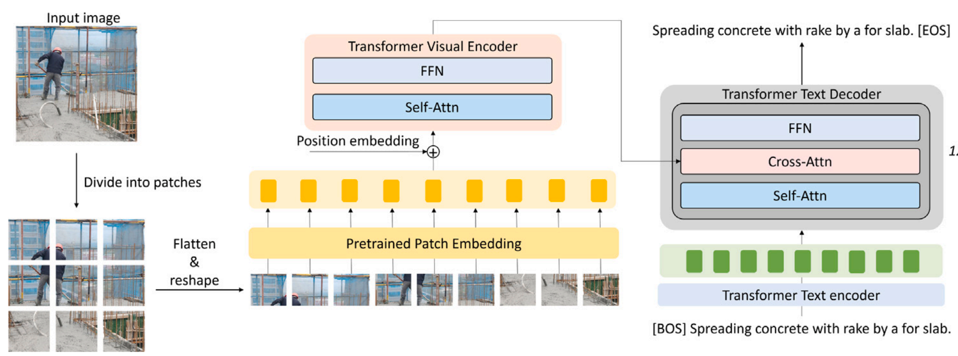

Fig. 4. Illustration of the baseline vision-language transformer model based on a pretrained mPLUG.

3.1. Vision-language transformer model architecture

For image captioning tasks, the Vision-Language Transformer (VL) model architecture combines a pre-trained Vision Transformer (ViT) architecture as an encoder for image feature representation with a separate language generation decoder to generate captions. The choice of ViT is inspired by [52,58,64] because this approach can be directly applied to a pure transformer to process images as sequences of patches, rather than the traditional reliance on regional operation in CNNs. ViT represents the simplest yet effective architecture for a VL model, employing the transformer module to extract and process visual features instead of separate deep visual embedder and regional operations, such as those found in convolutional visual embedding networks like Faster R-CNN and ResNets. As shown in Fig. 4, the ViT architecture consists of patch embedding, position embedding, and Transformer encoder layers. The input image is divided into patches, which are projected into a vector space called embeddings. Each position of image patches is preserved through position embedding. The ViT encoder layers allow the model to weigh the importance of different patches through the self-attention mechanism and then generate feature representations for each image patch.

To accomplish the captioning task, the presented model builds upon a modified and fine-tuned version of the mPLUG model [58]. The mPLUG is a vision-language Transformer, which is pretrained on large amounts of image-text paired multimodal datasets. By concatenating visual and linguistic features as input, this Vision-Language Transformer model architecture enables self-attention to detect alignments between visual and linguistic features. Furthermore, the cross-attention mechanism facilitates cross-modal interaction, conducting independent multi-modal fusion to address the information asymmetry problem effectively. For fine-tuning the mPLUG model, a visual transformer encoder converts raw image data into a sequence of embeddings, which are then used in a transformer decoder with cross-attention layers for visual-aware text generation in a sequence-to-sequence learning approach. Fig. 4 shows the architecture of the baseline vision-language transformer model, which is built upon a pretrained mPLUG. For the captioning task in this paper, this model is modified and fined-tuned through a process which is described in the following sections.

3.2. Pseudo Region features (PR)

In the ViT model architecture, a visual encoder extracts grid features about all patch information to cover the image as shown in Fig. 4. However, directly operating on flattened grid features can lead to a loss of spatial information [64]. To alleviate this problem and avoid the explicit use of an object detector, the region features are presented by embedding the region information of grid features based on multiple centroids of image patch clusters, inspired by recent works [64,65], which implicitly model auxiliary information into the model for more profound communications. The output of the mPLUG visual encoder, represented as a sequence of embeddings $\{v_0, v_1, \ldots, v_m\}$, where m denotes the number of grids, is utilized. This is then passed through the pseudo-region feature mechanism. The similarity between each grid feature $\{v_i\}$ and K learnable cluster centers $\{c_k\}$ is computed using a dot product function similar to Zeng et al. [64]. To map each grid feature with the nth cluster, the following equation is employed:

$$a_{i,n} = \frac{e^{v_i c_n^T + b_n}}{\sum_{k=1}^{N} e^{v_i c_k^T + b_k}} \tag{1}$$

where b_k is a trainable parameter associated with each cluster n, and $a_{i,n}$ ranges from 0 to 1. The closest region feature, r_n, is calculated as:

$$r_n = \| \sum_{i=1}^{m} a_{i,n}(v_i - \tilde{c}_n) \| \tag{2}$$

where $\tilde{c}_n$ is a learnable parameter of the same size as c_n and r_n, for each grid is normalized and concatenated with the output of the visual encoder before being fed to the decoder. Fig. 5 provides an illustration of how pseudo-region features are incorporated into our model, compensating for the loss of spatial information caused by flattening operation in the Transformer architecture. As shown later (Section 5.3), this step enriches model's ability to understand visual regions (i.e., spatial information).

3.3. High-level knowledge Pre-training (HP)

In this work, we develop a novel pre-training strategy called high-level knowledge pre-training (HP) to enhance the quality of generated

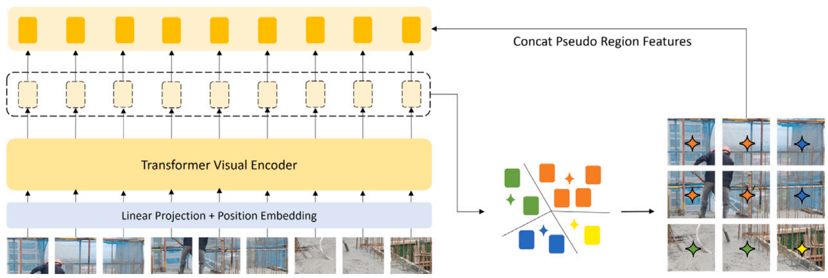

Fig. 5. Illustration of Pseudo Region features (PR) mechanism in VisualSiteDiary.

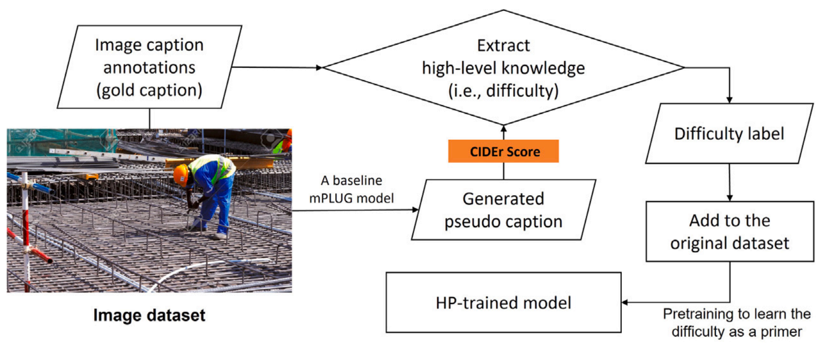

Fig. 6. Illustration of High-level knowledge Pre-training (HP).

captions. Pre-training provides a good initialization point [51,66], enabling the model to benefit from the knowledge gained during pre-training [67] and as such we use that to predict the overall difficulty of each input data image. As such, HP works as a primer for the model and makes the model well-prepared for the final image captioning fine-tuning step, by leveraging the pre-learned overall difficulty knowledge [68]. Fig. 6 illustrates the process.

The level of difficulty in creating image captions is evaluated based on the CIDEr score. To gauge the 'difficulty', first, the captions generated by the mPLUG baseline model is compared with the gold captions on the dataset. Then, we establish four labels: 'very easy', 'easy', 'hard', and 'very hard' based on the CIDEr quartile scores of our training data. With these newly assigned labels on our dataset, we (pre)train our model to learn the difficulty level associated with input images for five epochs. Through the HP process, the pre-trained large VL model can discern the complexity of each construction scene image, allowing the model to allocate greater consideration to images presenting higher difficulty levels during the caption generation process (i.e., fine-tuning), in contrast to images with relatively lower complexity.

3.4. Fine-tuning and accommodating user preferences

During the fine-tuning process, all parameters of a model are fine-tuned from pre-trained weights. As shown later, this allows the model to build upon its pre-trained knowledge and gains a better understanding of visual features and context, resulting in performance improvement [51,66,67].

VisualSiteDiary is fine-tuned for 20 epochs. VisualSiteDiary (default) is trained on both caption styles. To accommodate user preferences (compact style and detailed style), we provide two different fine-tuned models (VisualSiteDiary (compact) and VisualSiteDiary (detailed)) on each caption style separately. The motivation behind providing two different fine-tuned models is that users might want to receive captions tailored to their writing style or documentation requirements, enhancing the flexibility and usability of our image captioning model.

4. Experimental and data setup

4.1. Dataset

Our dataset is a combination of four existing image datasets and new image descriptions are annotated by the authors. The first dataset, Alberta Construction Image Data Set (ACID) [13], contained 4000 images of ten different types of construction equipment: excavator, compactor, dozer, grader, dump truck, concrete mixer truck, wheel loader, backhoe loader, tower crane, and mobile crane. Second dataset is from Liu et al. [12] which provides the image captioning dataset for five activity categories with a total of 5751 images collected from building construction sites: cart transportation, masonry work, rebar work, plastering, and tiling. Third dataset Zhai et al. [54] was created a total of 5211 images for captioning unsafe behavior of workers under five different construction objects: rebar, concrete, scaffold, wooden template, and bricks. The SODA dataset [69], the last dataset used, consists of 19,864 images for 15 construction objects: person, vest, helmet, board, wood, rebar, brick, scaffold, handcart, cutter, electric box, hopper, hook, fence, and slogan, is adopted for data augmentation using pseudo captioning.

Note, the ACID dataset mainly focuses on site work and substructure activities involving machinery, whereas the other three datasets [12,54,69] primarily cover building superstructure activities. We combined these datasets into one comprehensive dataset with our new image captions to evaluate the image captioning model's performance across various scenarios.

4.2. Data cleaning, reorganization and groundtruth development

Generating reasonable image descriptions DCR use-case requires a well-defined linguistic schema. For construction image captioning tasks, previous research [12,13,54] formalized captions with 'subject (machinery or workers)', 'action', 'object (e.g., soil, concrete, and brick)' and 'supplementary attributes' (e.g., count, safety gears). For instance, examples from these datasets include captions like 'a backhoe

Dataset	Subject	Object	Action	Attributes	Work Purpose
ACID [13]	Excavator, Wheel loader, Grader, Compactor, Dump truck, etc.	Soil, Ground, Surface, Gravel, Dirt, Rock, etc.	Excavating, Grading, Leveling, Stripping, Compacting, etc.	Count, Weather	Foundation, Earthwork, Site preparation, Underground utilities, etc.
Construction Activity [12]	Worker, Masonry worker, Carpenter, etc.	Brick, Rebar, Ceramic tile, Plaster	Laying, Tying, Finishing, Installing, Plastering, etc.	Count, Color, Safety gears	Masonry, Column, Slab, Floor, Wall, Exterior, Interior finish, etc.
Construction unsafe behavior [54]	Worker, Masonry worker, Carpenter, etc.	Brick, Rebar, Concrete, Scaffold, Wooden template	Laying, Tying, Pouring, Finishing, Installing, Framing, etc.	Count, Color, Safety gears	Masonry, Column, Slab, Floor, Wall, Exterior, Interior finish, etc.

Fig. 7. Revised linguistic concepts of the main scene elements on the three existing construction image captioning datasets. We newly incorporated "work purpose" to elaborate "what" activity information in the scene.

loader is pushing soil', 'a worker is laying a ceramic tile', and 'a concrete worker with a hard hat is pouring concrete' from [12,13,54], respectively.

However, this linguistic schema does not fully address the specific work items –*work purpose*– (e.g., leveling for foundation, pouring concrete for deck, and tiling for interior wall finish). Since including all significant activity objectives (i.e., 'what' information) that occur on the daily work and gathered visual information is significant for DCRs, we reorganized the linguistic schema referring to two real-world commercial building projects' daily reports and activity descriptions from 35 commercial building projects' master schedules. We introduce the concept of 'work purpose' (Fig. 7) and subsequently create new image descriptions (Fig. 8). Additionally, instead of simply replacing verbs with synonyms, we developed two distinct types of image descriptions, shown in Fig. 8: (1) a compact version for daily reports and updates, and (2) a detailed version that provides elaborated information on actions and safety.

Another concern lies in the similarity between images in the original training and validation set (see Fig. 9), where they share the same actions, poses, and jobsite environments. This redundancy in the training and validation dataset can lead to overfitting and may not validate the model performance when it applies to real-world problems. To mitigate this problem, we re-split training and validation images, considering their different scenes and jobsite environments.

Table 1 shows the skewed distribution of the captions for different categories in each original dataset inferred by counting their unigrams. As seen, the word frequencies are not equally distributed, especially masonry work in construction activities and reinforced in construction unsafe behavior, due to many redundant images by similar video files (Fig. 9). To reduce redundant images (same action, view, and jobsite environments) and balance the original datasets, we manually select 964 and 1762 images from [12,54] under different scenes as evenly as possible. Next, the dataset is augmented with additional images sourced from the SODA [69] object detection dataset using pseudo captioning techniques. This augmentation process enriches our dataset with a wider variety of construction scenes, as explained in Section 4.3.

4.3. Data augmentation using pseudo captioning

Because creating groundtruth data from construction scenes is time-changing and complicated, a high variety of scenes in the training data is required to ensure the model performance on unseen images. To augment the scenes without additional labor work, pseudo captioning is introduced to automatically obtain captions of new images from [69] by fine-tuning VisualSiteDiary using the new dataset. Cho et al. [70] used pseudo captions to develop an open-vocabulary object detector, distilling knowledge of the well-trained image captioning model. Combining with the common pseudo labeling, where a confidence threshold is used to filter predictions, our pseudo captioning exploits image-level information to decide if predicted captions should be added to the dataset.

Most computer vision solutions in the AEC industry were developed before recent vision-language models, and as such. more datasets for object detection [69,71] are available and contain images of various scenes. Taking advantage of the collected images and annotations, the fine-tuned model can make image-sentence pair of unseen scenes, and image-level information indicating which objects exist in an image is extracted to filter out wrong predictions. By adding such data without manually creating captions, the visual representation of objects is aimed to be enhanced. Fig. 10 illustrates data augmentation using pseudo captioning. Pseudo-captions are those that only mention objects existing in images but not necessarily all of them in the detection annotations. The augmented dataset will then be used to again fine-tune the model. The whole process can be performed iteratively so that the model matures on its performance.

As described previously, our VSD dataset was created to compensate for the previous dataset's problems with manual annotations. Since the original dataset is not equally distributed, the dataset is augmented with pseudo-captions as an automatic method that efficiently expands the coverage of the dataset. Using pseudo captioning, 1089 images from SODA [69] dataset for construction object detection are selected to mainly improve the limited scenes of superstructure activities. Table 2 shows the statistics of our final VSD dataset.

4.4. Evaluation metrics

We evaluate the performance of VisualSiteDiary using five widely used image captioning validation metrics: BLEU [72], ROUGE [73], METEOR [74], CIDEr [75], and SPICE [76]. These metrics help assess the quality and relevance of the model-generated captions by comparing them to reference captions created by human annotators.

4.4.1. BLEU

BLEU (Bilingual Evaluation Understudy) [72] is a metric originally developed for machine translation tasks, and it has been adapted for evaluating image captioning. BLEU quantifies the overlap of contiguous sequences of n words between the generated captions and the reference captions. It measures the precision of n-grams in the generated caption compared to reference captions, indicating how much the words in the machine-generated caption appear in the human reference captions. The BLEU score is expressed as a percentage, where 0 indicates no similarity between the generated captions and the reference captions. To compute the BLEU score, a brevity penalty is incorporated to penalize shorter generated captions and avoid favoring brevity:

$$\text{Brevity penalty (BP)} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}, \tag{3}$$

Scene	Image	Compact style (caption 1)	Detailed style (caption 2)	Original captions from [9,10,13]
Foundation	ACID [13] 	A trench box installation with an excavator for foundation	Lifting and moving a trench box with a wheel excavator for foundation	• Excavator is loading wooden framework • An excavator is lifing a steel beam using its bucket.
	ACID [13] 	Excavation for leveling foundation with two dump trucks and an excavator.	Dumping gravels into a dump truck with an excavator while another dump truck is waiting for leveling foundation.	• An excavator is loading two dump trucks. • An excavator is dumping soil into a dump truck and another dump truck is waiting to be loaded.
Rebar work	Construction unsafe behavior [54] 	Tying rebar by a group of workers for slab	Continued tying rebar by a group of workers with hard hats for slab	• a group of steel workers are binding steel bars • a group of steel workers were tying steel bars. • a group of steel workers are tying steel bars. • a group of steel workers are standing tying the bars. • a group of steel men with hard hats are standing tying steel bars.
Concrete work	Construction unsafe behavior [54] 	Pouring concrete by two workers for slab	Pouring concrete by two workers wearing hard hats for slab	• two concrete workers were pouring concrete. • two concrete workers with hard hats were pouring concrete. • two concrete workers in hard hats were pouring the concrete. • two concrete workers with helmets were pouring concrete. • two concrete workers in helmets were pouring the concrete.
Plastering	Construction Activity [12] 	Interior wall finish with plastering by a worker	Interior brick wall finish with plastering by a worker without hard hat information	• a man wearing a hat is plastering for a brick wall • a worker wearing a hat is plastering for a block wall • a worker in a hat is plastering for a brick wall • a man in a hat is plastering for a brick wall • a worker in a hat is plastering for a block wall

Subject | Purpose | Attribute (count)
Action | Object | Attribute (safety information)

Fig. 8. Revised caption examples by indicating work task purposes.

Fig. 9. Examples of train and validation dataset. (a): training and validation set from [12] has very similar actions and jobsite environment. (b): our new train and validation split considering diverse actions and jobsite environments to prevent the overfitting problem of the original dataset.

Table 1
The number of unigrams for different categories in each image captioning dataset.

Dataset	Category	Keywords (#)	Total number
ACID [13]	Excavator	excavator (1610), excavators (201)	1811
	Compactor	compactor (802), compactors (65)	867
	Dozer	dozer (665), dozers (40), bulldozer (65), bulldozers (3)	773
	Grader	grader (959), graders (24)	983
	Dump truck	dump (1985), dump-trucks (4), dump-truck (4), dumptrucks (1), dumptruck (1)	1995
	Concrete mixer truck	mixer (445), mixers (1)	446
	Wheel loader	wheel (1431), wheel-loader (2), wheel_loader (2), wheel-loaders (1), wheelloader (1)	1437
	Backhoe loader	backhoe (1053), backhoe-loader (2)	1055
	Tower crane	tower (246), tower-crane (1)	247
	Mobile crane	mobile (762), mobile-crane (1), mobile-cranes (1)	764
Construction activities [12]	Cart transporting	cart (372), carts (11)	383
	Masonry work	brick (13 509), bricklaying (51), bricks (763), block (8938), blocks (139)	23,400
	Rebar work	steel (701), bar (54), bars (708)	1463
	Plastering	plastering (1125), plastered (130)	1255
	Tiling	tile (1087), tiles (758), tile-laying (10), tiling (17)	1872
Construction unsafe behavior [54]	Reinforced	rebar (2), steel (12 798), bars (5654), bar (17)	18,471
	Concrete	concreter (825), concrete (7202), concreters (273)	8300
	Scaffold	scaffold (6477), scaffolder (1239), scaffolding (1976), scaffolders (1168), scaffolds (15)	10,875
	Wooden template	wooden (110), formwork (5872), forms (66), form (4)	6052
	Bricks	bricklayers (942), bricks (4580), brick (1572), bricklayer (1558)	8652

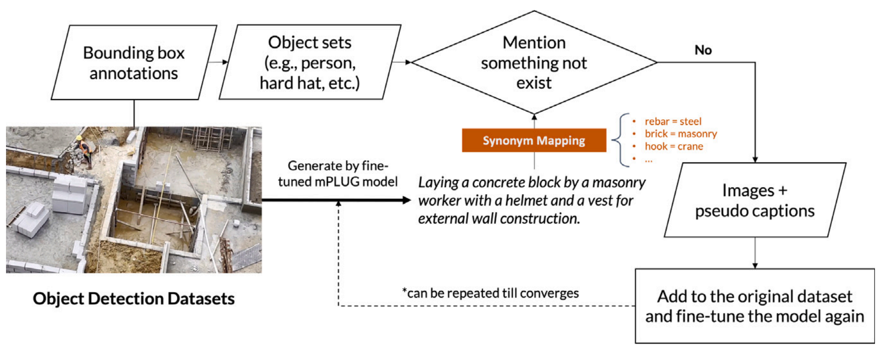

Fig. 10. Illustration of Pseudo captioning to augment training data.

Table 2
The comprehensive statistics of our dataset.

Our dataset		Our image #	Our caption #
ACID [13]	Train	3200	6400
	Val	800	1600
Construction activities [12]	Train	783	1566
	Val	181	362
Construction unsafe behavior [54]	Train	1562	3124
	Val	200	400
Object detection dataset [69]	Train	861	1722
	Val	228	456
Total	Train	6406	12 812
	Val	1409	2818

where c is the length of the generated caption, r is the reference corpus length. Then, the BLEU score is calculated as:

$$\mathrm{BLEU} = \mathrm{BP} \times \exp\left(\frac{1}{n}\sum_{i=1}^{n}\log(precision_i)\right), \tag{4}$$

where n is set as 4 in the baseline.

4.4.2. ROUGE_L

Similar to BLEU, ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [73] measures the overlap of n-grams in the human reference captions that appear in the machine-generated caption. ROUGE_L is a variant of the ROUGE metric that focuses on the longest common subsequence (LCS) between the generated and reference captions. The idea behind ROUGE-L is to emphasize the longest matching sequence of words (not necessarily consecutive), as it often represents the core meaning of the generated caption compared to simple n-gram matching. By considering the LCS, ROUGE-L addresses the limitation of standard ROUGE, which can be sensitive to the order of words and may not fully capture the semantic similarity between captions. The ROUGE scores also range from 0% to 100%, where a higher score indicates better similarity between the generated and reference captions.

$$\mathrm{ROUGE\_L} = \frac{(1+\beta^2)\mathrm{R_{LCS}}\mathrm{P_{LCS}}}{\mathrm{R_{LCS}}+\beta^2\mathrm{P_{LCS}}}, \tag{5}$$

where $\mathrm{R_{LCS}}$ and $\mathrm{P_{LCS}}$ are recall and precision of LCS, respectively. The parameter β is typically set to 1.2 [62].

$$\mathrm{R_{LCS}} = \frac{\mathrm{LCS(X,Y)}}{m} \quad \text{and} \quad \mathrm{P_{LCS}} = \frac{\mathrm{LCS(X,Y)}}{n}, \tag{6}$$

where m is the length of the reference caption, X, and n is the length of the generated caption, Y.

4.4.3. METEOR

METEOR (Metric for Evaluation of Translation with Explicit Ordering) [74] calculates the number of exact word matches and partial matches (i.e., alignment), allowing for the inclusion of synonyms and paraphrases. The alignment between the generated captions and reference captions is computed while minimizing the number of chunks by matching exact words word by word using WordNet synonyms [77], stem tokens, and paraphrases. A chunk is a set of consecutive words in an alignment. Ranging from 0% to 100%, a higher METEOR score indicates better similarity between the generated and reference captions. The METEOR score is calculated as follows:

$$\mathrm{METEOR} = (1-\mathrm{Penalty}) \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\alpha\mathrm{Precision} + (1-\alpha)\mathrm{Recall}} \tag{7}$$

$$\text{Penalty} = \gamma \times \left(\frac{\#\text{ of chunks}}{\#\text{ of unigrams matched}} \right)^{\beta}, \tag{8}$$

where α is a parameter for controlling relative weights of precision and recall. The default value of α is set as 0.9. The parameter β is for controlling the shape of the penalty as a function of fragmentation. The default value of β is 3. The parameter γ is a relative weight assigned to the fragmentation penalty. The default value of γ is set as 0.5.

$$\text{Precision} = \frac{m}{w_g} \quad \text{and} \quad \text{Recall} = \frac{m}{w_r}, \tag{9}$$

where m is the number of matched tokens (unigrams) between the generated caption and the reference caption. w_g is the number of words in the generated caption, while w_r is the number of words in the reference caption.

4.4.4. SPICE

While traditional evaluation metrics primarily focus on n-gram overlap for linguistic tasks, SPICE (Semantic Propositional Image Caption Evaluation) [76] leverages the semantic structure of scene descriptions to assess the semantic quality of generated captions. To generate semantic scene graphs, reference and machine-generated captions are parsed through dependency parse trees that encode objects, attributes, and relations. The similarity of generated and reference scene graphs is based on the F1-score over matching tuples in the generated and reference scene graphs. The use of WordNet [77] synonymy, following the approach of METEOR, allows SPICE to consider synonymous expressions while evaluating the captions. The SPICE score is reported on a scale from 0% to 100%, with 0% indicating no semantic similarity and 100% representing a perfect semantic alignment between the generated and reference captions.

4.4.5. CIDEr

CIDEr (Consensus-based Image Description Evaluation) [75] is designed to consider consensus among multiple reference captions by employing a Term Frequency Inverse Document Frequency (TF-IDF) weighting for each n-gram across the dataset. The core idea behind the CIDEr metric is that n-grams that frequently appear across all images in the dataset should be assigned lower weights since they are likely to be less informative (e.g., ‘a’ or ‘the’). The occurrence of an n-gram w_k in a reference sentence s_{ij} is represented by $h_k(s_{ij})$, while the occurrence in a generated sentence c_i is denoted by $h_k(c_i)$. The TF-IDF weighting $g_k(s_{ij})$ for each n-gram w_k is computed as follows:

$$g_k(s_{ij}) = \underbrace{\frac{h_k(s_{ij})}{\sum_{w_l \in \Omega} h_l(s_{ij})}}_{\text{TF of each } n\text{-gram } w_k} + \underbrace{\log\left(\frac{|I|}{\sum_{I_p \in I} \min(1, \sum_q h_k(s_{pq}))} \right)}_{\text{the rarity of } w_k \text{ using its IDF}}, \tag{10}$$

where Ω is the vocabulary containing all n-grams and I is the total number of images in the dataset.

The CIDEr_n score for n-grams of length n is calculated by computing the average cosine similarity between the generated sentence (c_i) and the reference sentences ($S_i = \{s_{i1}, \ldots, s_{im}\}$) for image I_i:

$$\text{CIDEr}_n(c_i, S_i) = \frac{1}{m} \sum_j \frac{g^n(c_i) \cdot g^n(s_{ij})}{\|g^n(c_i)\| \, \|g^n(s_{ij})\|}, \tag{11}$$

where g^n is a vector formed by g_k corresponding to all n-grams of length n and $\|g^n\|$ is the magnitude of the vector g^n.

$$\text{CIDEr}(c_i, S_i) = \frac{1}{N} \sum_{n=1}^{N} \text{CIDEr}_n(c_i, S_i), \tag{12}$$

where N is set as 4.

The CIDEr metric utilizes TF-IDF weighting to determine the importance of individual n-grams based on their frequency across the dataset, allowing it not to be bound to a specific range. Typically, CIDEr scores are positive values, with higher scores indicating better agreement between the generated captions and the reference captions. To enhance its robustness, this paper adopts CIDEr-D as a variant of the CIDEr metric. CIDEr-D encourages models to produce diverse and distinct captions for a given image by making two key modifications. First, stemming is removed, ensuring that the correct forms of words are utilized rather than mapping singular and plural forms and different tenses of verbs to the same token. Second, a Gaussian penalty is introduced, which mitigates cases in the basic CIDEr that yield higher scores when words of higher confidence are repeatedly used in lengthy sentences. The changes in CIDEr-D result in the following equation:

$$\text{CIDEr-D}_n(c_i, S_i) = \frac{10}{m} \sum_j e^{\frac{-(l(c_i)-l(s_{ij}))^2}{2\sigma^2}} \times \frac{\min(g^n(c_i), g^n(s_{ij})) \cdot g^n(s_{ij})}{\|g^n(c_i)\| \, \|g^n(s_{ij})\|}, \tag{13}$$

where $l(c_i)$ and $l(s_{ij})$ are the length of generated and reference sentences, respectively. σ is typically set as 6. The final CIDEr-D is calculated in a similar manner to CIDEr.

$$\text{CIDEr-D}(c_i, S_i) = \frac{1}{N} \sum_{n=1}^{N} \text{CIDEr-D}_n(c_i, S_i), \tag{14}$$

where N is set as 4.

4.5. Model setup

The visual transformer encoder of VisualSiteDiary is initiated by mPLUG [58] as the backbone of VisualSiteDiary. Following the setting of mPLUG, the text encoder is initiated by the first 6 layers of the BERT_{base} [51], and the text decoder is initialized using the last 6 layers of the BERT_{base}. For the pseudo region feature layer, the number of cluster centers K is set as 4. For high-level knowledge pretraining, we (pre)trained our model for 5 epochs with a learning rate of 1e−5 and batch size of 64 using AdamW [78], which is a variant of the popular Adam optimization by decoupling the weight decay from the gradient-based update. Next, VisualSiteDiary is fine-tuned with cross-entropy loss for 20 epochs with a learning rate of 1e−5 and a batch size of 64 using the AdamW optimizer with a weight decay of 0.02. The resolution for all input images is set to 384 × 384. Additionally, the maximum length of the generated captions is set as 25. All experiments are conducted on a single A100 GPU and with five random seeds, with average and standard deviation reported.

5. Results and discussion

In this section, we examine the quality of our new VSD dataset and the effects of the design decisions that distinguish VisualSiteDiary from baseline mPLUG through a number of analyses and ablation studies. Additionally, two real-world applications are presented: (1) automated DCR creation, and (2) text-to-image retrieval for multimedia content management to highlight the practical utility of VisualSiteDiary.

5.1. Analysis of dataset quality

For superstructure activities, since the original datasets [12,54] adopt a more simple linguistic schema and contain many consecutive images, the predicted captions might not include sufficient information for daily construction reports, and the models could suffer from unseen images. That is, the model can be vulnerable to overfitting based on such datasets. To verify this, we separately fine-tune the mPLUG model using the original datasets and VSD dataset. The fine-tuned models are then evaluated with their corresponding validation sets. Instead of comparing the overall performance, the metric values of each image are calculated one by one, and the result is summarized to present the performance distribution. In Fig. 11, the metric values of each image are divided by their maximum value, and the number of images is accumulated as the percentage along with ascending metric values. For the ACTV [12] and SAFE [54] datasets, Fig. 11(a) shows that more than half of the images reach the highest score as well as the

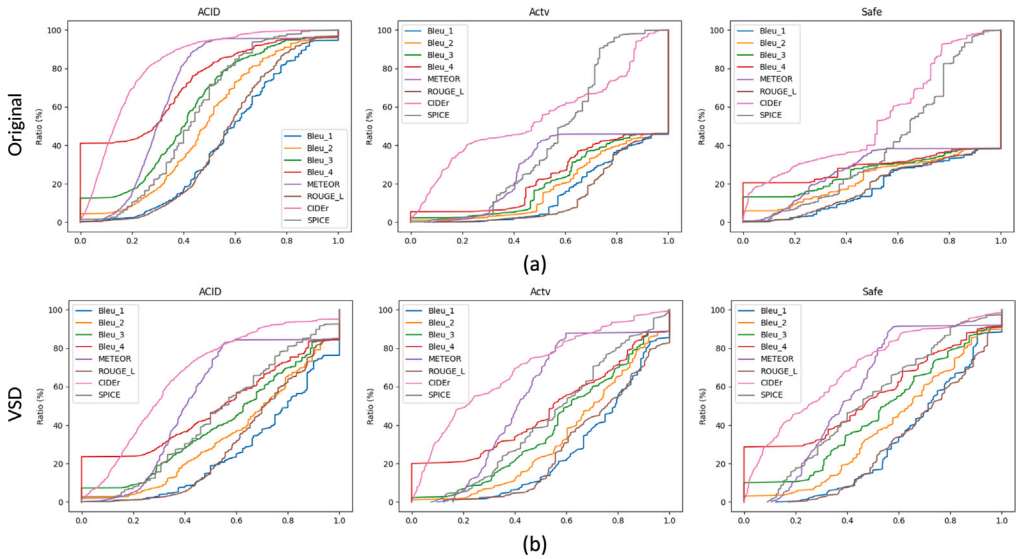

Fig. 11. The performance distribution based the original [12,13,54] and VSD validation sets. x: the min–max normalized metric values; y: accumulated percentage. Figure (a) shows a sharp rise at the start or the end of the curve, demonstrating vulnerability to overfitting when using the original dataset. In contrast, Figure (b) presents smoother changes in the evaluation metric values on our VSD dataset, indicating mitigation of the extreme distribution observed in the original dataset and enhanced quality.

Table 3
The performance improvement of tower crane images on the ACID validation set.

Training dataset	BLEU@1	BLEU@2	BLEU@3	BLEU@4	METEOR	ROUGE_L	CIDEr	SPICE
Using images from [12,13,54] (Before augmentation)	67.4	58.8	51.4	42.2	37.5	65.8	203.5	65.9
Augmentation with images from [69] (VSD dataset)	69.8	60.5	53.2	44.7	38.8	67.5	252.8	67.3

sharply climbing curves at the start and end. The situation indicates that the difficulties of the validation images are not evenly distributed. By removing some consecutive images and re-formulating captions to satisfy the DCR application, VSD dataset leads to a better-distributed performance on each validation image. Fig. 11(b) presents more smooth changes in the metric values, and the percentage of the extreme values becomes less among the whole datasets.

On the contrary, for substructure activities, the ratio of very low scores in the performance distribution based on the original ACID dataset [13] is much higher than the other two datasets in Fig. 11(a). The extremely high score distribution in the ACTV [12] and SAFE dataset [54] is due to the adopted synonym substitution strategy for augmenting the caption number up to 5 times greater and redundant scenes across training and validation sets. The ACID dataset was annotated by engineers using different personal styles and as such, the dataset does not contain suggestions of objects and corresponding actions. While the captioning principle makes the ACID's captions more random and various, the training image samples are not enough and cause a poorer performance on the ACID dataset. In addition to addressing extreme distribution, VSD dataset also mitigates the negative impact of different captioning styles, revising them to be focused on the important work purpose for further DCR application. As demonstrated in Fig. 11(b), the ratio of very low performance decreases, and the distributions of the three datasets become more similar to each other.

Additionally, through the augmentation from the SODA dataset [69] using pseudo captions, additional images of tower cranes are added to alleviate the imbalance of the number of objects that could cause biases. Using all the 94 images of tower cranes in the ACID validation set, the models based on the dataset with and without augmentation are evaluated. Table 3 presents that all the metrics are improved about 2% after augmentation. By introducing more scenes without additional labor work but with image-level information to filter out wrong predictions, the training dataset can be successfully enhanced, and the feasibility of incorporating other available datasets from vision tasks is validated.

5.2. Overall performance of VisualSiteDiary

Table 4 shows the overall performance of VisualSiteDiary along with the state-of-the-art baselines on VSD dataset. We observe that mPLUG is noticeably better than the other two baseline models (CLIP [57] and GiT [80]), presumably due to its effective pre-training task. Based on this result, mPLUG forms the backbone of VisualSiteDiary, and PR and HP are applied on top to further improve the overall performance. By comparing VisualSiteDiary and mPLUG from Table 4, we observe that VisualSiteDiary outperforms the mPLUG baseline in all five metrics and 0.7 points on the overall average. We also conduct a significance test comparing VisualSiteDiary with mPLUG for a more robust evaluation. Since a single performance score is insufficient to validate the effectiveness of the model, we follow [81] and apply the test with a significance level of 0.01. We observe that the p-value is less than 0.01 for the overall score validating the effectiveness of VisualSiteDiary and performance improvement over the baseline model (Details are provided in Appendix C).

5.3. Ablation studies and qualitative analyses

In Sections 3.2 and 3.3, we described two methods of PR and HP, to improve the performance of VisualSiteDiary. In this section, we demonstrate their advantage through ablation studies and qualitative analyses.

5.3.1. Quantitative results

Ablation studies are conducted with and without each method, and the results are summarized in Table 4. We observe that both PR and HP improve the mPLUG baseline model (avg. score from 110.3 to 110.7 for PR and from 110.3 to 110.8 for HP) and, when combined, further boost the performance (up to 111.0), verifying the effectiveness and the orthogonality of the two methods. We analyze the model-generated captions in detail, in the next section.

Table 4
Overall, ablation, and style-specific results of VisualSiteDiary. All experiments are the average of 5 runs with random seeds. Best performance marked in bold.

Dataset: VSD	BLEU@4	METEOR	ROUGE_L	CIDEr	SPICE	Avg.
1. Overall results						
• CLIP [79]	44.7 (±0.2)	32.9 (±0.4)	60.3 (±0.4)	235.9 (±1.0)	42.9 (±0.7)	83.3 (±0.5)
• GiT [80]	44.8 (±0.2)	32.8 (±0.3)	60.1 (±0.5)	237.2 (±1.3)	43.4 (±0.4)	83.2 (±0.6)
• mPLUG [58]	57.2 (±0.4)	39.6 (±0.1)	70.4 (±0.2)	331.5 (±2.3)	52.8 (±0.3)	110.3 (±0.6)
• **VisualSiteDiary (p-value < 0.01)**	**58.0 (±0.3)**	**39.9 (±0.1)**	**70.6 (±0.2)**	**333.7 (±3.0)**	**53.0 (±0.2)**	**111.0 (±0.7)**
2. Ablation results						
• VisualSiteDiary	58.0 (±0.3)	39.9 (±0.1)	70.6 (±0.2)	333.7 (±3.0)	53.0 (±0.2)	110.0 (±0.7)
• VisualSiteDiary w/o PR	57.6 (±0.3)	39.8 (±0.2)	70.5 (±0.3)	333.4 (±2.2)	52.9 (±0.2)	110.8 (±0.6)
• VisualSiteDiary w/o HP	57.8 (±0.2)	39.9 (±0.1)	70.5 (±0.2)	332.5 (±2.5)	52.9 (±0.2)	110.7 (±0.6)
• VisualSiteDiary w/o PR & HP	57.2 (±0.4)	39.6 (±0.1)	70.4 (±0.2)	331.5 (±2.3)	52.8 (±0.3)	110.3 (±0.6)

3. Style-specific results	Avg. word count	Avg. score (compact)	Avg.score (detailed)
• VisualSiteDiary (Default)	9.9	98.3 (±0.4)	105.6 (±0.6)
• VisualSiteDiary-Compact	8.6	**113.4 (±0.2)**	74.4 (±0.6)
• VisualSiteDiary-Detailed	11.2	72.5 (±0.5)	**122.8 (±0.3)**

5.3.2. Qualitative results

To delve into the effectiveness of VisualSiteDiary in utilizing image information with the support of PR and HP, we compare the results between VisualSiteDiary and the vanilla mPLUG. The qualitative comparisons reveal noticeable differences in the generated captions (Detailed examples of qualitative results are provided in Fig. A.15 (Appendix A)). For example, the vanilla mPLUG created 'a wheel loader arrives on site'. caption, not reflecting the fact that the image has two different machines, a wheel loader and a dump truck. However, VisualSiteDiary generated a more accurate caption, 'earthwork with a wheel loader and a dump truck for site work', correctly incorporating information about both pieces of equipment. Moreover, in cases where multiple machines were present, such as an excavator and two dozers, VisualSiteDiary accurately presented image explanations with all equipment information, while the baseline mPLUG model generated 'two excavators' only (Fig. A.15). Similarly, the vanilla mPLUG failed to decipher concrete vibrator information, while our model generated the caption with concrete vibrator information. These results indicate that incorporating regional features implicitly through our PR method allows VisualSiteDiary better to recognize accurate objects than the baseline mPLUG model.

Another advantage of VisualSiteDiary lies in its good proficiency in handling complex and challenging images that are difficult to interpret. Considering that the images acquired from original datasets encompass diverse weather and lighting conditions, as well as various construction stages, such variability may pose challenges when directly applying state-of-the-art approaches [82,83]. For instance, as shown in Fig. A.15, the gold caption contains four heavy equipment such as 'with two wheel loaders and two excavators for foundation'. However, the actual objects in the image under cloudy weather and overexposed in the brightest parts of the image are two excavators, one wheel loader, and a dump truck. Despite this complexity and difficulty in deciphering the scene, VisualSiteDiary generated a correct caption that accounted for the dump truck, understanding the complex content of the image. Since HP pre-training enables the model to learn the difficulty level of captioning tasks, VisualSiteDiary can leverage the learned difficulty information and generate more precise captions, even in challenging scenarios.

By incorporating regional information and learning the level of difficulty in each image, VisualSiteDiary achieves superior results, generating more contextually relevant captions. In contrast, the baseline mPLUG struggles to comprehend visual information accurately. For instance, the caption generated by the baseline mPLUG 'a worker wearing a hard hat is standing on scaffold' misses the crucial action of 'tying scaffold'. On the other hand, VisualSiteDiary can decipher images with precise information based on actions, purposes, and objects. For example, instead of merely deciphering the purpose as 'for wall', VisualSiteDiary understands the work purpose as 'for exterior wall'. Similarly, VisualSiteDiary can detail action information, generating captions like 'fixing and tying' rather than a simplistic 'tying', and 'applying mortar to ceramic tile' rather than 'laying ceramic tile'. Therefore, through these qualitative comparisons, we demonstrate the effective ability of VisualSiteDiary in generating image captions that fully contain 'what' and 'how' information using the PR and HP strategy for the DCR creation.

5.4. Variable style-caption generation

Undoubtedly, superintendents and project managers have different preferred writing styles. Therefore, providing different caption styles based on the user's preference is beneficial and crucial to satisfy various users. Based on this motivation, we provide two additional versions of VisualSiteDiary (VisualSiteDiary-Compact and VisualSiteDiary-Detailed) to accommodate a broader range of users who may prefer different writing styles. Each version is fine-tuned using its corresponding style of gold captions. Note that this is possible due to the unique characteristic (having two distinct styles of gold captions per image) of the VSD dataset. Using these two variants together allows users to pick and use the one corresponding to their preferred caption style, as a result providing broader options to the users.

VisualSiteDiary is analyzed and the two variants in detail, and the results are summarized in the third section of Table 4 (note that the original VisualSiteDiary is denoted by VisualSiteDiary (Default)). Here, the default model shows low scores when the gold captions are restricted to a specific style (98.3 w.r.t. compact gold captions and 105.6 w.r.t. detailed gold captions). This is because the default model, trained with both styles of captions, generates an arbitrary style of caption given an input image (i.e., VisualSiteDiary (Default) generates compact captions for some images, detailed captions for some others, and mixed-style captions for the rest). Due to this unpredictable behavior, the default VisualSiteDiary is inappropriate for generating captions with the style set by the user. On the other hand, the two variants are each trained with only a single style of gold captions, thereby able to generate a fixed style of captions. The results showing much higher scores on their corresponding caption styles (113.4 w.r.t compact gold captions for VisualSiteDiary-Compact and 122.8 w.r.t detailed gold captions for VisualSiteDiary-Detailed) and the average word counts of the three models (VisualSiteDiary-Compact (8.6) < VisualSiteDiary (Default) (9.9) < VisualSiteDiary-Detailed (11.2)) verify this assumption.

As a result, by utilizing VisualSiteDiary-Compact and VisualSiteDiary-Detailed together, the users can generate their preferred style of captions for any input image, a practical advantage over other image captioning models.

5.5. Application 1: VisualSiteDiary *for automated daily construction report (DCR)*

Today, reality mapping which automatically maps images and videoframes against floor plans, organizes them over a project timeline in

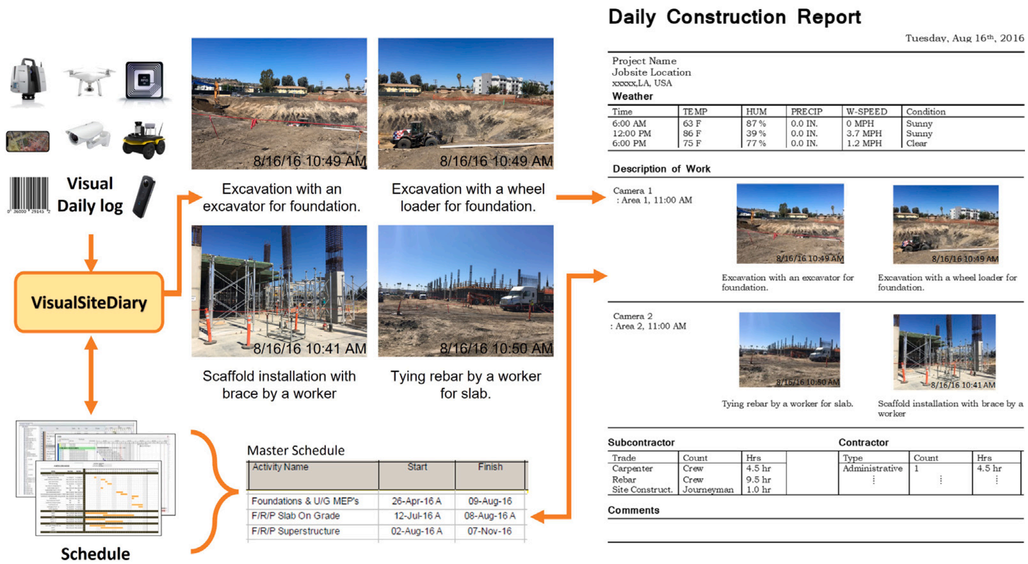

Fig. 12. Automated DCR generation using VisualSiteDiary. Image descriptions are generated based on the compact style caption. The captured on-site visuals contain timestamps (i.e., 'when') and offer spatial context (i.e., 'where') such as Level 1 Zone A via reality mapping.

a measurable form, provides the 'when' and 'where' context of daily construction photologs. To create DCRs using this data, two challenges should have been addressed [6,7]: providing (1) 'what' specific activities take place and (2) 'how' the site has evolved and changed over time. In this work, we demonstrated the potential of integrating VisualSiteDiary with modern reality mapping solutions. By combining CV and NLP techniques, VisualSiteDiary allows the automatic generation of descriptive textual information ('what' and 'how' information) for images captured on the construction site. When jobsite photos are captured and uploaded to our VisualSiteDiary, the uploaded visuals contain timestamps (i.e., 'when' information). On top of this, the reality mapping system assists in identifying spatial context (i.e., 'where' information) such as Level 1 Zone A based on the visual information. The generated descriptions can be integrated with the timestamp and spatial information, yielding the automation of DCR creation. Fig. 12 shows an example of VisualSiteDiary to create DCR using real-world project images. Also, the generated captions can be compared with the project schedule as a means of simple progress tracking, since they include significant activities that align with the master schedule activity descriptions.

Another line of our method application for automated DCR involves using time-lapse images from installed closed-circuit television (CCTV) cameras on the jobsite. To demonstrate the capabilities of VisualSiteDiary, we use 12 images from one-day time-lapse images. Given that the time-lapse images are stored in the system based on a temporal sequence, a part of generated captions is repetitive if the jobsite environment is not much changed. To handle these images, we first generate each caption on each time-lapse image via VisualSiteDiary, and then implement a BERT-based simple extractive summarization model [84] to compress the important activities. As shown in Fig. 13, we can extract the significant transitions of activities and textual descriptions from the time-lapse images. Therefore, VisualSiteDiary proves its ability to generate human-readable descriptions for fully automated DCRs purely based on visual information. This approach not only streamlines the reporting process but also improves the comprehensiveness and efficiency of daily reporting in construction. Furthermore, the images employed in the generation of those DCRs (Figs. 12 and 13) are external images, not included during the training or validation phase. This inference serves to illustrate the robustness and adaptability of VisualSiteDiary.

5.6. Application 2: Image-text retrieval for multimedia content management

VisualSiteDiary enables the integration of images and text-based information. By generating text-based image descriptions, VisualSiteDiary extends visual-based content to the multimedia dataset (images with human-readable descriptions) and enables users to do a proper text-based search using natural language or queries. Table 5 shows the result of text-to-image retrieval, where users input object-level text to get relevant images. For instance, using 'masonry' as a query retrieves relevant images with 87.6% of recall value. These text-based image searches are helpful to a number of applications such as identifying most active areas of work through text queries about certain work packages; and reviewing the latest work for payment application.

To demonstrate the performance for this use-case, we compare the object-level recall value and the precision of generated captions between VisualSiteDiary (ours) and the baseline mPLUG. The captions were created more accurately containing important object information with VisualSiteDiary (89.5% precision) than the baseline mPLUG (88.9% precision). With more accurate captions, VisualSiteDiary conducts the text-to-image retrieval task better with a 78.4% recall value than the baseline mPLUG (77.5% recall value).

To conduct an interactive search and exploration within the image-text paired dataset, images are deployed into a vector space by extracting visual features from the VisualSiteDiary visual encoder. Fig. 14 shows an example vector space of the visual features extracted from the VisualSiteDiary visual encoder. The vector space provides insights into visual similarities, including human-readable captions and high-level category information. When a user selects one of their images, our system automatically visualizes and retrieves related images based on their distance (i.e., similarity) in a vector space. For example, we observe that images of rebar activity are situated close to scaffold activity images, while concrete activity images are located nearby formwork activities.

This demonstrates the expanded capabilities of VisualSiteDiary beyond image captioning and the effectiveness of VisualSiteDiary, which empowers users to utilize multimodal datasets intuitively and facilitates seamless coordination among different project stakeholders, such as owners, contractors, and subcontractors, for effective project control and administration. Therefore, this integration of visual and text-based information leads to an efficient search system that fosters human–computer interaction and content management.

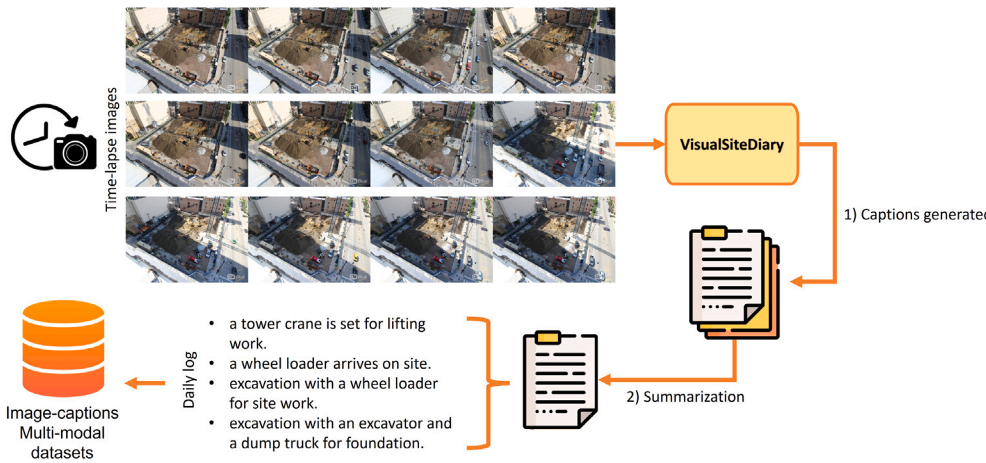

Fig. 13. An example of employing time-lapse images for DCR and daily log generation using VisualSiteDiary (compact style version).

Table 5
The comparison of object-level recall and accuracy in text-to-image retrieval between our VisualSiteDiary and the baseline mPLUG model. TP = True Positive.

Object	# items	Recall of objects (%)		# of TP	Precision of captions (%)	
		VisualSiteDiary	mPLUG		VisualSiteDiary	mPLUG
Dump truck	377	90.5	92.0	202	85.1	86.2
Excavator	367	89.6	88.8	176	94.9	96.0
Dozer	145	72.4	69.7	61	88.5	85.2
Wheel loader	260	81.9	80.8	136	80.9	76.8
Backhoe loader	197	88.3	90.4	101	87.1	88.2
Mixer truck	105	94.3	96.2	56	89.3	85.0
Compactor	174	86.2	86.2	79	96.2	96.2
Tower crane	179	77.7	52.5	89	79.8	87.3
Mobile crane	144	83.3	87.5	62	96.8	88.7
Worker	1110	95.7	96.4	551	96.4	96.2
Helmet	546	34.6	37.2	199	95.0	96.7
Vest	142	28.2	26.1	50	80.0	77.1
Rebar	400	86.5	81.5	198	87.4	78.7
Concrete	427	66.5	71.5	160	93.1	87.8
Masonry	251	87.6	80.2	126	88.1	83.7
Plastering	106	100.0	100.0	68	77.9	88.3
Tiling	94	66.0	70.2	33	93.9	94.3
Scaffold	186	52.2	57.0	68	77.9	85.1
Formwork	118	85.6	61.0	75	68.0	69.8
Total	**5322**	**78.4**	**77.5**	**2485**	**89.5**	**88.9**

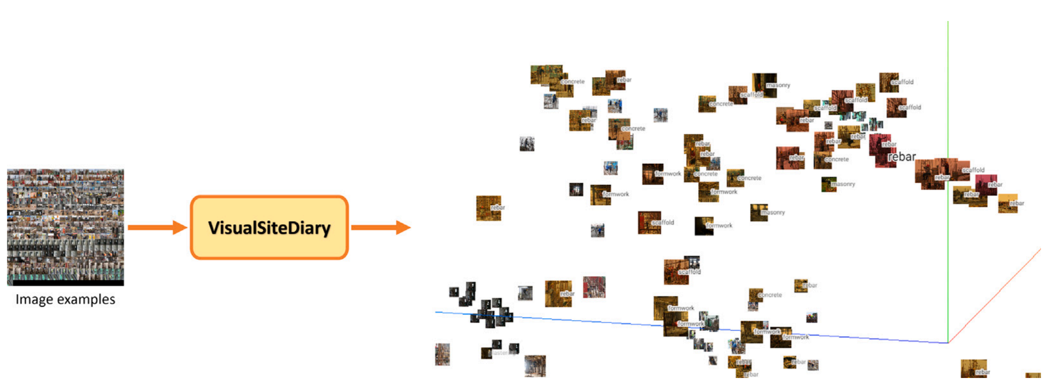

Fig. 14. When a user selects one of their images, the multimedia content management system retrieves related images automatically based on their distance in a vector space, enabling efficient search and fostering human–computer interaction.

5.7. Strengths and limitations

This section highlights the notable strengths of the proposed methodology while also acknowledging potential limitations that may influence its applicability and effectiveness in real-world construction scenarios. Further research and development efforts could focus on addressing these limitations to enhance the robustness and versatility of the proposed approach.

Strengths:

- Innovative methodology: Introducing novel techniques like pseudo-region features and high-level knowledge pretraining enhances the Vision-Language (VL) Transformer model, VisualSiteDiary, for construction image captioning, replacing costly convolutional operations.
- VSD dataset coverage & caption quality: VSD is the first well-balanced construction image-caption dataset encompassing sitework, substructure activity, and superstructure activity. It ensures the robustness of VisualSiteDiary and adaptability for DCR creation and text-to-image retrieval.
- Comprehensive evaluation: The proposed model, VisualSiteDiary, is thoroughly evaluated using diverse metrics. VisualSiteDiary achieves state-of-the-art performance across all five metrics, compared to baseline models, demonstrating its effectiveness and robustness. On top of this, the ability of VisualSiteDiary for the DCR creation application is demonstrated with unseen real-world project images (external construction site images).
- Accommodating user preferences: The VSD dataset includes two caption styles (compact and detailed), enabling separate fine-tuning of models for each style. This enhances applicability and usability in practical construction applications, catering to user preferences.

Limitations:

- Inherent challenges in fine-tuning: While the VSD dataset covers diverse construction scenarios, inherent limitations in fine-tuning for entirely new scenes such as interior furniture and miscellaneous interior accessories and MEP activities may affect the quality of captions in the construction contexts.
- Need for real-world survey: Although validated with VSD for the model performance and external data for demonstrating applications, further real-world assessment via surveys to superintendents and managers may provide deeper insights into VisualSiteDiary's capabilities.
- Discrete controllability: Fine-tuning strategies separately accommodate discrete user preferences, potentially limiting controllability when users seek intermediate options between caption styles (e.g., 0.3 of detail level in the range of compact (level 0) to detail (level 1)).

6. Conclusions

This paper proposes a novel vision transformer-based model, called VisualSiteDiary, that automatically generates image content descriptions on various scenarios, unlocking a new way of two real-world applications: (1) automated daily construction report system and (2) text-to-image retrieval for multimedia content management. In contrast to previous approaches, VisualSiteDiary directly employs a pure transformer to process images as sequences of patches, rather than the traditional reliance on CNNs with bounding boxes. More specifically, we incorporate three key approaches using a pretrained large vision-language model. First, we effectively embedded pseudo visual region features (PR) into the model, addressing the loss of spatial information caused by the flattening operation in the ViT architecture. Second, we developed a novel pre-training strategy called high-level knowledge pre-training (HP) to leverage the knowledge of difficulty levels during pre-training. Lastly, we provide two distinct fine-tuning processes that allow customization according to user preferences or documentation requirements. As a result, VisualSiteDiary produces superior-quality captions compared to other state-of-the-art image captioning models with 111.0 points on average of five evaluation metrics, enhancing flexibility and usability.

To validate the method, we create a comprehensive image captioning dataset called VSD, based on our new linguistic schema, which includes important activity purposes (i.e., work purposes) for DCR generation. The enhanced quality of image captioning datasets is demonstrated by conducting a thorough comparison of the performance distribution between our VSD dataset and the three original image captioning datasets from [12,13,54]. In addition, the applications of the presented model for the purposes of generating DCRs and image-text retrieval were discussed in detail. Specifically, two external construction project images are used to demonstrate the model's performance in DCR generation. The experimental results show that the VisualSiteDiary achieves 89.5% precision in image-text retrieval tasks, further validating the application of the presented method for multimedia content management tasks in construction. For future work, the capabilities of the VisualSiteDiary model can be expanded to include other applications, such as mapping look-ahead tasks from a project schedule, incorporating semantic segmentation for computer vision progress monitoring, and coupling with generative chatbot systems such as ChatGPT. Additionally, the image captioning dataset can be further augmented to enhance its richness and diversity. Additional practical investigations with site superintendents and field managers and by using practical Key Performance Indicators (KPIs) such as cost and schedule performance can also help with offering a deeper understanding of the return-on-investment of systems that build on VisualSiteDiary's model.

CRediT authorship contribution statement

Yoonhwa Jung: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ikhyun Cho:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Shun-Hsiang Hsu:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Software, Validation, Visualization. **Mani Golparvar-Fard:** Supervision, Project administration, Conceptualization, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yoonhwa Jung reports financial support was provided by National Science Foundation.

Data availability

I have shared the link to our data.

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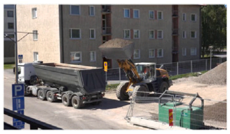
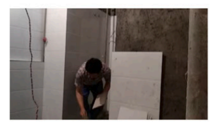
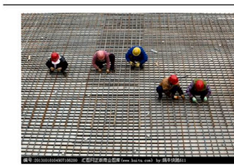
Image	Captioning result	Image	Captioning result
	Baseline (mPLUG): **a wheel loader** arrives on site. **Ours (VisualSiteDiary): earthwork** with a wheel loader and **a dump truck for site work** Gold caption: **earthwork for site** preparation with a wheel loader and a **dump truck.**		Baseline (mPLUG): a worker wearing a hard hat is standing on scaffold. **Ours (VisualSiteDiary): fixing and tying scaffold** by a worker wearing a hard hat Gold caption: **tying scaffold** by a worker wearing a hard hat.
	Baseline (mPLUG): excavation with **two excavators for site preparation**. **Ours (VisualSiteDiary):** excavation with **an excavator and two dozers for site work.** Gold caption: earthwork for site preparation with **an excavator and two dozers.**		Baseline (mPLUG): pouring concrete by two workers for slab. **Ours (VisualSiteDiary): compacting** concrete **with vibrator** by two workers for slab Gold caption: **vibrating** concrete **with vibrator** by three workers for slab
	Baseline (mPLUG): excavating and loading soil and rocks into **a dump truck with an excavator** for earthwork. **Ours (VisualSiteDiary): excavating and loading** soil and rocks into **two dump trucks with two excavators for foundation.** Gold caption: loading soil and rocks **with two wheel loaders and two excavators for foundation.**		Baseline (mPLUG): brick masonry by a worker **for wall.** **Ours (VisualSiteDiary):** brick masonry by a worker **for exterior wall** Gold caption: brick masonry installation by a worker **for exterior wall**
	Baseline (mPLUG): tying rebar by a worker for wall. **Ours (VisualSiteDiary): installing ceramic tile for the wall by a worker** Gold caption: **ceramic tile installation for exterior wall by a worker** on the scaffold		Baseline (mPLUG): laying ceramic tile by a worker without a helmet for interior wall finish. **Ours (VisualSiteDiary): applying mortar** to ceramic tile by a worker without a helmet for interior wall finish **Gold caption: applying mortar** to ceramic tile by a worker without a helmet for interior wall finish
	Baseline (mPLUG): tying rebar by a group of workers with helmets for slab. **Ours (VisualSiteDiary):** tying rebar by a group of workers with helmets for slab. Gold caption: tying rebar by a group of workers with helmets for slab.		Baseline (mPLUG): interior wall finish with plastering by a worker. **Ours (VisualSiteDiary):** interior wall finish with plastering by a worker Gold caption: interior wall finish with plastering by a worker

Fig. A.15. Qualitative results comparing our VisualSiteDiary results with baseline mPLUG results.

Appendix A. Examples of generated captions via VisualSiteDiary

VisualSiteDiary is presented for automated DCR creation. To demonstrate the effectiveness of the support of the PR and HP methods, 10 images from the bottom 100 were randomly selected by sorting them in descending order based on the CIDEr value. This experiment allows us to demonstrate performance on the most difficult images to create captions. Fig. A.15 shows qualitative results of comparing the generated captions for these images for both VisualSiteDiary and the baseline mPLUG model. Our demo[2] provides several live inference examples.

Appendix B. Original image-caption datasets in construction

Existing construction image-caption pair datasets [12,13,54] are limited in scope, covering only specific scenarios such as earthwork or superstructure activities separately as discussed in Section 4.2. Consequently, research based on these datasets lacks comprehensive validation across diverse construction contexts. Additionally, the division of training/validation sets within these datasets often results in redundancy, with same poses, actions, and jobsite environments depicted. To address these limitations, we combined existing image-caption datasets to encompass a broader range of construction activities and restructured training/validation sets to eliminate redundancy and balance dataset composition, resulting in no redundancy across the training and validation sets. Moreover, unlike other datasets which simply substitute verbs with synonyms, VSD includes two distinct types of image descriptions (compact and detailed), enhancing usability and accommodating user preferences. Although the original datasets appear extensive in terms of images and captions (See Table B.6), their redundancy both in images and captions and lack of real-world validation are notable limitations.

Table B.6
The statistics of original construction image-caption datasets.

Dataset		Image #	Caption #
ACID [13]	Train	3200	6735
	Val	800	1493
Construction activities [12]	Train	4601	23 005
	Val	1150	5750
Construction unsafe behavior [54]	Train	3648	18 240
	Val	1563	7815

Appendix C. Significance test

Deep learning models, with their numerous hyperparameters and non-convexity, converge based on random initialization during training. The significance test aims to determine the superior and statistically significant algorithm by analyzing empirical validation score distributions. Following the approach from [81], we compared our method (VisualSiteDiary) with a baseline model (mPLUG). The input of this significance test consists of three components (results from method

[2] Demo is made available at https://github.com/joonv2/VisualSiteDiary/tree/demo.

A, results from method B, and a confidence level) to evaluate whether one method is "stochastically greater" than the other, with details provided in [81]. The test yields a minimum epsilon value, interpreted to ascertain the significance of the results as follows:

- $\epsilon > 0.5$: We cannot say A is better than B.
- $\epsilon \leq 0.5$: A is better than B.
- $\epsilon = 0$: A is stochastically dominant over B.

With a confidence level of 0.99 (equivalent to a significance level of 0.01), the significance test conducted on our model yielded minimum ϵ values of 0. This outcome indicates that VisualSiteDiary exhibits stochastic superiority over the mPLUG baseline at a confidence level of 0.99, affirming the efficacy of our approach with a statistical significance of $p < 0.01$.

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